

Yusen Su

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Nationality: Chinese (PRC)

Research Profile

Researcher in program analysis and formal verification specializing in abstract interpretation, LLVM-based static analysis, and bounded model checking. Builds practical abstract domains and verification pipelines in the **SeaHorn** framework; industry experience verifying OpenHarmony OS and AWS SDK for C production codebases.

Education

- 2020–2026 **Ph.D., Electrical and Computer Engineering**, *University of Waterloo*,
(Defended) Faculty of Engineering (Adviser: Arie Gurfinkel) Grade: 87.75/100
Dissertation *Scalable Program Analysis: Abstract Interpretation Techniques and Practical Applications*
○ Designed novel abstract domains and compositional pre-analysis pipelines for LLVM-based verification, improving precision and scalability of memory safety proofs, relational invariant inference, and taint analysis.
- 2018–2020 **M.Sc., Computer Science**, *New York University*,
Courant Institute of Mathematical Sciences (Adviser: Thomas Wies) GPA: 3.90/4.0
Dissertation *Data Flow Refinement Type Inference Tool: DRIFT²*
- 2015–2016 **B.Sc., Computer Science**, *University of Minnesota - Twin Cities*,
College of Science and Engineering GPA: 3.66/4.0
- 2012–2014 **B.Sc., CS and Technology (Dual degree)**, *University of Electronic Science and Technology of China*, School of Computer Science and Engineering GPA: 3.77/4.0

Professional Experience

- Sep 2023 - **Associate Researcher (Part-time)**, *Huawei Technologies Canada*, Waterloo, Canada
Apr 2025 ○ Designed a symbolic-execution-guided test-case generation tool for C/C++ APIs: compiled targets to ELF binaries and used **angr** to drive execution toward uncovered CFG branches
○ Automated test synthesis via **Z3** path-constraint solving to generate inputs for **RapidCheck** property-based test generators
○ Evaluated on a ~1K-LOC OpenHarmony OS utility library and ~10K-LOC Huawei internal libraries; generated ~200 test cases and improved branch coverage from 40% to 80%; tool deployed in the team's CI pipeline
- May 2023 - **Associate Researcher (Intern)**, *Huawei Technologies Canada*, Waterloo, Canada
Aug 2023 ○ Extended **SeaBMC** with C standard library models (string operations) for memory safety verification; applied a divide-and-conquer verification strategy across industrial C codebases (networking, database, utility modules)—proving API-level properties and propagating results as assumptions to reduce BMC proof obligations
○ Identified buffer overflow vulnerabilities in database and networking components missed by existing unit tests
- Sep–Dec 22 **Teaching Assistant**, *University of Waterloo*, Waterloo, Canada
+ Sep–Dec 21 ECE 351 - Compilers (Undergraduate)
○ Held weekly office hours, graded assignments and exams, and provided feedback on parser and code-generation projects

- Jan–May 20 **Recitation Leader and Grader**, *New York University*, New York City, USA
+May–Aug 19 CSCI-GA.2110 - Programming Languages (Graduate)
 - Led weekly recitation sessions and graded assignments covering type systems, lambda calculus, and operational semantics
Sep–Nov 2017 **Web Developer Intern**, *Harbin Institute of Technology*, Harbin, China
Ground Station TM/TC System - **Mun**
 - Self-designed a TCP-WebSocket proxy client and built the **Tornado** web server
 - Designed the web server architecture and completed the integration testing

Research Artifacts

- 2021–Present **Crab – Abstract Interpretation Library for LLVM**, *University of Waterloo*
 - Contributed 4 novel abstract domains and a heap-aware taint analysis engine
 - **Numerical domain** for memory-access validation: evaluated on 2,620 API-level programs, demonstrating its good balance between precision and performance — *VSTTE 2025*
 - **Memory domain** for object invariants: evaluated on 109 GNU Coreutils programs; as a pre-analysis pass, outperforms standalone bounded model checking — *VMCAI 2025*
 - **Heap-aware taint analysis**: evaluated on 183 programs across various benchmark suites (Amazon IONC, CURL, Coreutils) — *submitted to SAS 2026*
2025–Present **SeaDSA – A Points-to Analysis for LLVM**, *University of Waterloo*
 - **Interval-aware DSA extension**: designed and implemented finer-grained object disjointness and pointer alias disambiguation; integrated with taint analysis, demonstrating precision improvements at comparable runtime — *submitted to SAS 2026*
2020–Present **SeaBMC – Bounded Model Checker for LLVM**, *University of Waterloo*
 - Integrated **Crab** as pre-analysis passes in the BMC pipeline to filter provable assertions and speeds up both BMC encoding and SMT solver performance.
2019–2020 **DRIFT² – Data Flow Refinement Type Inference Tool**, *New York University*
 - Implemented and experimentally evaluated an abstract interpretation-based refinement type inference algorithm; demonstrated precision and efficiency advantages over Liquid Types-based verification tools — *POPL 2021*

Technical Skills

- Languages C/C++, Python, OCaml, Rust, Java, Scheme
Formal Abstract Interpretation, Static Analysis, Symbolic Execution, Bounded Model Checking,
Methods Taint Analysis
Tools Z3 (API, tactic tuning), SeaHorn (SeaDSA, Clam/Crab, SeaBMC), CBMC, angr, KLEE
Compilers LLVM (IR, passes, transformations), Git, Linux/Unix systems
Spoken Mandarin (native), English (fluent), Spanish (beginner)

Publications

- Submitted **[SAS’26]** Y. Su, J. A. Navas, A. Gurfinkel. **Taint Analysis via Heap-Aware Propagation.**
2025 **[VSTTE’25]** Y. Su, J. A. Navas, A. Gurfinkel. **Template DBM: A New Weakly Relational Domain for Efficient Memory-Access Validation** *17th International Conference on Verified Software: Theories, Tools, and Experiments. Best Tool Paper Award*
2025 **[VMCAI’25]** Y. Su, J. A. Navas, A. Gurfinkel, I. Garcia-Contreras. **Automatic Inference of Relational Object Invariants.** *Verification, Model Checking, and Abstract Interpretation.* DOI: 10.1007/978-3-031-82700-6_10

- 2022 **[FMCAD'22]** S. Priya, Y. Su, Y. Bao, X. Zhou, Y. Vizel, A. Gurfinkel. **Bounded Model Checking for LLVM**. *22nd Formal Methods in Computer-Aided Design*. DOI: 10.34727/2022/ISBN.978-3-85448-053-2_28
- 2021 **[ATVA'21]** S. Priya, X. Zhou, Y. Su, Y. Vizel, Y. Bao, A. Gurfinkel. **Verifying Verified Code**. *Automated Technology for Verification and Analysis*. DOI: 10.1007/978-3-030-88885-5_13
- 2021 **[POPL'21]** Z. Pavlinovic, Y. Su, T. Wies. **Data flow refinement type inference**. *Proc. ACM Program. Lang.* DOI: 10.1145/3434300

Conference Service

- Subreviewer Tools and Algorithms for the Construction and Analysis of Systems (TACAS) '26, '23, '22; FMCAD '25, '22; International Symposium on Formal Methods (FM) '24, '23; NASA Formal Methods Symposium (NFM) '22
- AEC Computer Aided Verification (CAV) '26, '25; TACAS '26, '25
AEC = Artifact Evaluation Committee
- Volunteer CAV '24

Honors and Awards

- 2025 Best Tool Paper Award, *17th International Conference on Verified Software: Theories, Tools, and Experiments (VSTTE 2025)*
- 2020–present Graduate Research Studentship, University of Waterloo
- 2012–2014 National Endeavor Fellowship, University of Electronic Science and Technology of China

Leadership and Activities

- Summer 2013 **Rutgers China Bridge Program in Engineering**, *Rutgers University*,
Direct Leadership Team, UESTC